MASTER VOLUME · 10 COUNTY CIRCULATION DRAFT

SYNC / Green Code Master Volume for 10 Priority County Chapters

Dark-mode compilation for UNESCO / EPA circulation, preserving the eight-county pilot reference while extending the portfolio with Fresno and Santa Clara supplemental chapters.

COUNTY COUNT	PRESERVED PRIOR REFERENCE	BUDGET SUBTOTAL CARRIED FORWARD
10 county chapters	**8 county pilot PDF appended as Appendix A**	**$44.06M research · $60.29M full planning across the original eight-county cohort**

SUPPLEMENTAL COUNTY STATUS	CORE EVIDENTIARY RULE	TECHNICAL STANDARD REFERENCE
Fresno and Santa Clara added as pre-budget chapters sourced from the national-scale reference text	**Engineering, fiction, grant framing, and audit lanes remain separate**	**SYNC_GreenCode_Protocol_v2.1.pdf**

How this volume should be read: the original eight-county package remains the preserved reference foundation. This master volume does not overwrite it. Instead, it compiles a circulation-ready front section, adds two supplemental California county chapters, and appends the preserved reference PDF so reviewers can move from summary and governance framing into the underlying earlier deliverables without losing source continuity.

Budget caution: the only aggregated totals reported in this master volume are the already-established eight-county subtotals: **$44.06M** for combined research framing and **$60.29M** for combined full-planning envelopes. Fresno and Santa Clara are included as supplemental chapters but intentionally remain outside those totals until county-partner scoping occurs.

VECTOR A LEAD + LEAKAGE | VECTOR B COOLING + WUE | VECTOR C DEMAND SCHEDULING | UNESCO §§ 3.2 + 4.3 | DARK-MODE CIRCULATION SET | PRESERVED APPENDIX SOURCE

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EXECUTIVE SUMMARY

Portfolio logic across the ten-county volume

Original eight-county core:
Newark, Philadelphia, Bronx, Harris County, Cook County, Miami-Dade, Wayne County, and Los Angeles form the already-developed cohort with split deliverables and an existing eight-county reference PDF.

Supplemental expansion:
Fresno and Santa Clara extend the volume into rural/agricultural infrastructure and Silicon Valley data-center accountability without pretending that their budget framing is already fully negotiated.

Publication discipline:
Do not migrate fictional scenarios, modeled ROI, or unverified partner assumptions into pilot-design claims, grant budgets, or regulator-facing summaries.

What the master volume is not:
It is not a county-by-county guarantee of adoption, and it is not a substitute for partner MOUs, IRB review, or utility access agreements.

COUNTY ATLAS

Ten counties at a glance

COUNTY / SERVICE AREA	RESEARCH BUDGET	FULL PILOT ENVELOPE	AUDIT STATUS	EDITORIAL READ
1. Essex / Newark Service Area New Jersey	$4.93M	$6.50M	5 / 8 / 10 / 3	Most mature evidentiary base
2. Philadelphia County Pennsylvania	$4.42M	$6.30M	4 / 6 / 7 / 2	Strong candidate with legal/reporting relevance
3. Bronx County New York	$4.70M	$6.06M	4 / 4 / 8 / 1	High public-interest relevance
4. Harris County / Houston Texas	$5.57M	$9.58M	1 / 6 / 13 / 0	Institutionally complex
5. Cook County / Chicago Illinois	$4.87M	$6.23M	3 / 5 / 11 / 0	Technically interesting for Vector B
6. Miami-Dade County Florida	$6.54M	$8.71M	3 / 5 / 11 / 0	High-concept, high-risk advanced pilot
7. Wayne County / Detroit Michigan	$4.74M	$6.52M	4 / 5 / 8 / 1	Strong Vector A candidate
8. Los Angeles County California	$8.29M	$10.39M	3 / 5 / 9 / 0	Largest and costliest concept
9. Fresno County California	Pre-budget supplemental chapter	To be scoped after partner access	Provisional audit — scoring pending	Supplemental California field chapter
10. Santa Clara County California	Pre-budget supplemental chapter	To be scoped after operator commitments	Provisional audit — scoring pending	Supplemental Silicon Valley chapter

Audit status uses the prior split-package counts where available. Fresno and Santa Clara are marked provisional because they are newly integrated here from the national-scale reference text rather than a previously finalized county packet.

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COUNTY CHAPTERS

Compiled deliverables for all ten counties

Each county chapter below follows the same lane structure: real pilot design, clearly marked GSF-3 fiction, research-grant framing, and data-audit discipline. For the original eight-county cohort, these sections summarize and align the preserved reference material. For Fresno and Santa Clara, the sections act as supplemental chapters prepared from the national-scale reference text and kept pre-budget until county-specific partner scoping is complete.

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NEW JERSEY · MOST MATURE EVIDENTIARY BASE

Essex / Newark Service Area

County focus: Lead-service-line accountability, NRW reduction, acoustic leak detection, and privacy-preserving attestation in a utility with documented lead-crisis context.

Research	**$4.93M**
Full pilot	**$6.50M**
Audit	**5 / 8 / 10 / 3**

DELIVERABLE 1 · REAL PILOT DESIGN

Architecture lane: Vector A + C pilot with acoustic mesh, ZKP attestation, Vector A/C integration, range-proof reporting, one-DMA MVP appendix, and verification-latency target under four seconds.

Verified anchors usable in non-fiction: ~42,000 city-owned LSLs; NRW 20.3% (2022); BLL >5 µg/dL at 8.4% (2019) and 7.2% (2022); documented $120M replacement program; smart-meter infrastructure noted in prior package.

Success posture: Leak-detection speed +30%; ZKP false-positive under 1%; false-negative under 3%; NRW reduction at least 3 percentage points; publication only after governance review.

DELIVERABLE 2 · GSF-3 FICTION (EXPLICITLY SPECULATIVE)

Green Code Newark 2031 remains explicitly speculative: engineer Darnell, cryptographer Yuna, resident Marisol, and Commissioner Velez illustrate bounded system behavior without converting fiction into evidence.

DELIVERABLE 3 · RESEARCH GRANT FRAMING

NSF/EPA framing is already developed around Newark as the primary site with Camden comparison logic, RQ1–RQ5, SCADA + acoustic + smart-meter + BLL + work-order data, and a $4.93M research budget.

Recommended common data lanes: SCADA / historian access where available, acoustic or pressure telemetry, smart-meter or billing data where available, work orders, public-health data only under appropriate IRB and privacy pathways, and publication review before any public release.

DELIVERABLE 4 · DATA AUDIT

Retain the verified/partial/unverified/fiction separation already established in the split package. Do not move fictional BLL improvement or exact savings claims into the engineering lane.

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PENNSYLVANIA · STRONG CANDIDATE WITH LEGAL/REPORTING RELEVANCE

Philadelphia County

County focus: Consent-decree-aligned reporting, lead-service-line tracking, NRW reduction, and bounded exploratory campus or enterprise data-center measurement.

Research	**$4.42M**
Full pilot	**$6.30M**
Audit	**4 / 6 / 7 / 2**

DELIVERABLE 1 · REAL PILOT DESIGN

Architecture lane: Primary Vector A architecture with privacy-preserving compliance reporting and bounded data-center overlay; regulatory language softened to potentially acceptable rather than presumptively accepted.

Verified anchors usable in non-fiction: ~42,000 city-owned LSLs; NRW about 22%; EPA and PA DEP consent-decree context; dual Vector A + exploratory Vector B framing from the preserved 8-county reference.

Success posture: Demonstrate defensible attestations for compliance-facing metrics, faster leak identification, and cleaner source separation between utility operations and any cooling-use overlay.

DELIVERABLE 2 · GSF-3 FICTION (EXPLICITLY SPECULATIVE)

The 2031 scenario remains fiction only and cannot be cited as evidence for regulatory acceptance, citywide savings, or BLL change.

DELIVERABLE 3 · RESEARCH GRANT FRAMING

Proposal framing centers on dual-vector stress, partner engagement, audit-first publication, and research questions about compliance usability, public trust, and bounded pilot transferability.

Recommended common data lanes: SCADA / historian access where available, acoustic or pressure telemetry, smart-meter or billing data where available, work orders, public-health data only under appropriate IRB and privacy pathways, and publication review before any public release.

DELIVERABLE 4 · DATA AUDIT

Remove exact zone counts, exact post-pilot BLL projections, and any exact citywide savings figure from non-fiction lanes.

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NEW YORK · HIGH PUBLIC-INTEREST RELEVANCE

Bronx County

County focus: High-lead-exposure neighborhoods, private-side participation, and community-trust constraints for privacy-preserving infrastructure reporting.

Research	**$4.70M**
Full pilot	**$6.06M**
Audit	**4 / 4 / 8 / 1**

DELIVERABLE 1 · REAL PILOT DESIGN

Architecture lane: Vector A + C with community-advisory requirements, privacy-preserving enrollment logic, and strict separation between verified inventory and hypothetical household uptake.

Verified anchors usable in non-fiction: ~110,000–130,000 LSLs; childhood BLL >5 µg/dL reported around 8.4%; NYC LSLR budget about $1.3B; Bronx NRW around 18%.

Success posture: Demonstrate whether privacy-preserving attestation can improve participation and repair prioritization without overstating causal health effects.

DELIVERABLE 2 · GSF-3 FICTION (EXPLICITLY SPECULATIVE)

Any future participation-rate growth or dramatic exposure reductions stay fiction-bound unless later field evidence exists.

DELIVERABLE 3 · RESEARCH GRANT FRAMING

Research framing emphasizes equity, trust, privacy, and the operational question of whether cryptographic attestation can help target scarce replacement capacity.

Recommended common data lanes: SCADA / historian access where available, acoustic or pressure telemetry, smart-meter or billing data where available, work orders, public-health data only under appropriate IRB and privacy pathways, and publication review before any public release.

DELIVERABLE 4 · DATA AUDIT

Keep the “highest in nation” language qualified; all future participation and health-improvement numbers remain scenario-only.

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TEXAS · INSTITUTIONALLY COMPLEX

Harris County / Houston

County focus: Multi-utility coordination across selected MUDs, leakage detection, data-center water-use reporting, and exploratory ERCOT-linked demand scheduling.

Research	**$5.57M**
Full pilot	**$9.58M**
Audit	**1 / 6 / 13 / 0**

DELIVERABLE 1 · REAL PILOT DESIGN

Architecture lane: Vector B strongest, with bounded A and C overlays. Revised language treats ZKP as a proposed coordination mechanism for 8–12 participating MUDs rather than a solved regional system.

Verified anchors usable in non-fiction: 100+ MUDs; Houston-area data-center capacity roughly 350–480 MW in the preserved package; county NRW around 15.2%; large utility coordination challenge is real.

Success posture: Show whether shared attestations can coordinate narrow pilot partners, reduce reporting friction, and support leakage plus cooling-use accountability without overclaiming regional coverage.

DELIVERABLE 2 · GSF-3 FICTION (EXPLICITLY SPECULATIVE)

Any precise ERCOT demand-response revenue or region-wide ROI should stay scenario-level until real partner commitments exist.

DELIVERABLE 3 · RESEARCH GRANT FRAMING

Grant framing focuses on multi-operator governance, operator privacy, cooling-use disclosure, and resilience under hurricane and wet-bulb constraints.

Recommended common data lanes: SCADA / historian access where available, acoustic or pressure telemetry, smart-meter or billing data where available, work orders, public-health data only under appropriate IRB and privacy pathways, and publication review before any public release.

DELIVERABLE 4 · DATA AUDIT

Keep ROI labeled as an estimate and state explicitly that the pilot covers only a limited partner set, not the full county.

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ILLINOIS · TECHNICALLY INTERESTING FOR VECTOR B

Cook County / Chicago

County focus: Data-center WUE reporting, cooling optimization, and bounded waste-heat-recovery concepts in a major metro corridor with residual Vector A context.

Research	**$4.87M**
Full pilot	**$6.23M**
Audit	**3 / 5 / 11 / 0**

DELIVERABLE 1 · REAL PILOT DESIGN

Architecture lane: Vector B dominant. Revised summary explicitly avoids treating Vector C as a prioritized lane here, resolving the prior inconsistency.

Verified anchors usable in non-fiction: Chicago-metro data-center capacity around 450–520 MW; LSL count around 250,000–300,000; NRW roughly 20–24%.

Success posture: Test whether operator-private WUE attestations and bounded heat-recovery accounting can be made regulator-readable without exposing proprietary facility data.

DELIVERABLE 2 · GSF-3 FICTION (EXPLICITLY SPECULATIVE)

Heat-recovery offsets and aggressive WUE improvements remain modeled or fictional until site partners publish real measurements.

DELIVERABLE 3 · RESEARCH GRANT FRAMING

Research framing asks whether cryptographic reporting can reduce competitive disclosure barriers that currently block metro-scale operator cooperation.

Recommended common data lanes: SCADA / historian access where available, acoustic or pressure telemetry, smart-meter or billing data where available, work orders, public-health data only under appropriate IRB and privacy pathways, and publication review before any public release.

DELIVERABLE 4 · DATA AUDIT

Do not state waste-heat performance as established future output; keep it proposal-level.

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FLORIDA · HIGH-CONCEPT, HIGH-RISK ADVANCED PILOT

Miami-Dade County

County focus: Extreme-climate cooling, hurricane resilience, tropical infrastructure monitoring, and coastal water constraints.

Research	**$6.54M**
Full pilot	**$8.71M**
Audit	**3 / 5 / 11 / 0**

DELIVERABLE 1 · REAL PILOT DESIGN

Architecture lane: Vector B extreme-climate test case with resilience overlays and bounded Vector A context. Post-storm claims stay explicitly conditional.

Verified anchors usable in non-fiction: Data-center capacity around 120–160 MW; wet-bulb temperature around 24.5°C; NRW roughly 22–28%; LSL count roughly 90,000–120,000.

Success posture: Establish whether resilience-facing attestations can be generated rapidly enough to support infrastructure triage after storm events while staying privacy-preserving.

DELIVERABLE 2 · GSF-3 FICTION (EXPLICITLY SPECULATIVE)

Six-hour post-hurricane recovery or attestation claims remain fictional unless demonstrated in a real event or exercise.

DELIVERABLE 3 · RESEARCH GRANT FRAMING

Research framing emphasizes humidity-limited cooling physics, hurricane resilience, and the governance challenge of publishing performance without overselling certainty.

Recommended common data lanes: SCADA / historian access where available, acoustic or pressure telemetry, smart-meter or billing data where available, work orders, public-health data only under appropriate IRB and privacy pathways, and publication review before any public release.

DELIVERABLE 4 · DATA AUDIT

Flag the six-hour post-storm recovery claim as fiction and keep seawater-rejection efficacy unverified.

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MICHIGAN · STRONG VECTOR A CANDIDATE

Wayne County / Detroit

County focus: Lead legacy, cast-iron main failures, affordability pressure, and trust-sensitive participation in post-industrial neighborhoods.

Research	**$4.74M**
Full pilot	**$6.52M**
Audit	**4 / 5 / 8 / 1**

DELIVERABLE 1 · REAL PILOT DESIGN

Architecture lane: Vector A dominant with Vector C affordability framing; community-engagement-heavy pilot logic and conservative publication posture.

Verified anchors usable in non-fiction: ~90,000–100,000 LSLs; NRW around 23–26%; median pipe age around 67 years; proxy BLL figure around 6.8% in prior package.

Success posture: Determine whether cryptographic attestation can improve prioritization and accountability in a failure-prone legacy network without overstating household outcomes.

DELIVERABLE 2 · GSF-3 FICTION (EXPLICITLY SPECULATIVE)

Any private-side enrollment growth and household-credit effects remain scenario-only.

DELIVERABLE 3 · RESEARCH GRANT FRAMING

Research framing focuses on failure detection, prioritization, repair transparency, and whether privacy-preserving proofs can support contested infrastructure decisions.

Recommended common data lanes: SCADA / historian access where available, acoustic or pressure telemetry, smart-meter or billing data where available, work orders, public-health data only under appropriate IRB and privacy pathways, and publication review before any public release.

DELIVERABLE 4 · DATA AUDIT

Retain an even tone; avoid condescending phrasing and keep financial/social outcome claims in the scenario lane only.

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CALIFORNIA · LARGEST AND COSTLIEST CONCEPT

Los Angeles County

County focus: Large-scale data-center cooling, multi-agency water oversight, heat-stress governance, and large-utility reporting coordination.

Research	**$8.29M**
Full pilot	**$10.39M**
Audit	**3 / 5 / 9 / 0**

DELIVERABLE 1 · REAL PILOT DESIGN

Architecture lane: Vector B dominant with multi-agency coordination as the core challenge. Prior geography mixing with Santa Clara has been corrected; Los Angeles is treated as its own county chapter.

Verified anchors usable in non-fiction: Large data-center market noted; LADWP NRW roughly 10–12%; LSL count around 180,000+; water-stress context is material.

Success posture: Evaluate whether a regional operator coalition could share bounded cooling and water-use attestations without exposing proprietary data or collapsing agency boundaries.

DELIVERABLE 2 · GSF-3 FICTION (EXPLICITLY SPECULATIVE)

Any county-wide operator savings or fully regionalized response claims remain modeled until actual agency coordination exists.

DELIVERABLE 3 · RESEARCH GRANT FRAMING

Research framing focuses on scale, institutional complexity, and phased participation rather than county-wide immediate deployment.

Recommended common data lanes: SCADA / historian access where available, acoustic or pressure telemetry, smart-meter or billing data where available, work orders, public-health data only under appropriate IRB and privacy pathways, and publication review before any public release.

DELIVERABLE 4 · DATA AUDIT

Preserve the corrected geography and avoid importing Santa Clara-specific zoning or facility assumptions.

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CALIFORNIA · SUPPLEMENTAL CALIFORNIA FIELD CHAPTER

Fresno County

County focus: Rural and peri-urban leakage, canal-and-groundwater operations, drought-index correction, and agricultural interface monitoring.

Research	**Pre-budget supplemental chapter**
Full pilot	**To be scoped after partner access**
Audit	**Provisional audit — scoring pending**

DELIVERABLE 1 · REAL PILOT DESIGN

Architecture lane: Vector A + C field-heavy overlay: wellfields, canal SCADA bridge, urban district metered areas, acoustic mesh, and drought-index correction for emergency storage integration.

Verified anchors usable in non-fiction: Population 1,008,654; 47 major water systems; ~4,200 miles of pipe; NRW estimate 26.4%; Fresno Irrigation District / canal SCADA and wellfield context appear in the national-scale reference text.

Success posture: Use Fresno as a rural-to-urban stress test for whether attested leak reduction and demand coordination can be translated from fragmented systems into a publishable, audit-first pilot design.

DELIVERABLE 2 · GSF-3 FICTION (EXPLICITLY SPECULATIVE)

The Fresno fiction lane should remain clearly marked because the national-scale text contains modeled efficiency gains that are not yet county-verified.

DELIVERABLE 3 · RESEARCH GRANT FRAMING

Grant framing should ask whether a fragmented, agriculture-adjacent county can support privacy-preserving leak and allocation reporting across mixed public, rural, and irrigation actors. Budget is intentionally left unaggregated until county-partner scoping occurs.

Recommended common data lanes: SCADA / historian access where available, acoustic or pressure telemetry, smart-meter or billing data where available, work orders, public-health data only under appropriate IRB and privacy pathways, and publication review before any public release.

DELIVERABLE 4 · DATA AUDIT

Treat the NRW estimate, simulated efficiency gains, and ROI as reference-model values from the national-scale document unless Fresno utility partners validate them.

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CALIFORNIA · SUPPLEMENTAL SILICON VALLEY CHAPTER

Santa Clara County

County focus: Data-center cooling accountability, operator-private WUE reporting, cumulative-water range proofs, and Silicon Valley utility coordination.

Research	**Pre-budget supplemental chapter**
Full pilot	**To be scoped after operator commitments**
Audit	**Provisional audit — scoring pending**

DELIVERABLE 1 · REAL PILOT DESIGN

Architecture lane: Vector B + C overlay with county WUE thresholds, operator-private monthly readings, cumulative-water proofs, and bounded emergency-limit attestations rather than public disclosure of proprietary performance.

Verified anchors usable in non-fiction: SCVWD manages 10 reservoirs and 7 recharge ponds; serves roughly 2 million people; San Jose Water serves ~1 million; the national-scale reference states 340+ data centers representing 45% of California data-center capacity.

Success posture: Test whether county-scale cooling-water attestations can lower disclosure friction while producing regulator-readable, allocation-sensitive evidence in a high-density data-center market.

DELIVERABLE 2 · GSF-3 FICTION (EXPLICITLY SPECULATIVE)

Any dramatic county-wide water savings, thermal recovery, or ROI values remain modeled until operator data-sharing agreements and measured pilots exist.

DELIVERABLE 3 · RESEARCH GRANT FRAMING

Grant framing should focus on WUE thresholds, cumulative-water allocation proofs, public-versus-private utility coordination, and the governance challenge of making operator data auditable without revealing trade secrets. Budget remains pre-scoping.

Recommended common data lanes: SCADA / historian access where available, acoustic or pressure telemetry, smart-meter or billing data where available, work orders, public-health data only under appropriate IRB and privacy pathways, and publication review before any public release.

DELIVERABLE 4 · DATA AUDIT

Treat facility counts, WUE targets, and simulated gains as sourced from the national-scale reference text but not yet independently audited for a county-specific grant attachment.

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APPENDIX BRIDGE

Appendix A — preserved 8 county pilot reference

The pages following this bridge reproduce the earlier **8 county pilot** PDF as a preserved reference appendix. It remains the authoritative source for the earlier eight-county package structure, edits, and prior wording. This master volume is therefore additive and editorially corrective rather than destructive.

> **Appendix contents:** Essex / Newark, Philadelphia, Bronx, Harris County, Cook County, Miami-Dade, Wayne County, and Los Angeles, each with pilot design, GSF-3 fiction, grant framing, and data-audit lanes as previously developed.

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Priority Counties 4–10: Full Deliverable Package

One Sentence to Rule Them All (All Counties)

These four deliverables serve different functions and must remain evidentially separate: the pilot design is a proposed research pathway, the fiction is explicitly speculative, the grant framing is conditional on partner confirmation, and the data audit defines the boundary between verified fact, partial evidence, and fictional extrapolation.

PRIORITY COUNTY 4: Philadelphia, Pennsylvania

DELIVERABLE 1: Real Pilot Design

Philadelphia, Pennsylvania — Pilot Design

Why Philadelphia Is a Strong Pilot Candidate

Philadelphia is **one of the strongest pilot candidates** because:

Criterion	Philadelphia Advantage
Documented crisis	2019 lead crisis; ongoing EPA/PA DEP consent decree (2011)
Regulatory framework	Consent decree requires compliance reporting; PWD under court oversight
Scope manageable	City-owned LSL inventory documented at ~42,000; private-side requires sampling
Active program	PWD LSLR program with documented progress; capital budget committed
Data availability	EPA SDWIS, PA DEP records, PWD annual reports exist
Vector B component	Growing data center market (180–250 MW) enables dual-vector pilot

Note on LSL scope: Broader total service-line exposure is often estimated above the documented city-owned count, but private-side inventory remains incomplete. Pilot sizing must be treated as illustrative until field sampling is complete.

Pilot Scope

What This Pilot Tests

Component	Test Question
ZKP attestation schema	Can we prove LSL replacement compliance and NRW reduction without revealing private customer data?
Vector A + B integration	Does leak repair + data center cooling optimization produce measurable water savings?
Vector A + C integration	Does leak repair + smart scheduling interact meaningfully?
Consent decree reporting	Can ZKP reporting satisfy EPA consent decree requirements?
Acoustic mesh deployment	Can continuous monitoring reduce detection time in combined sewer areas?

What This Pilot Does NOT Test

Component	Reason
National scalability	Out of scope for single-utility pilot
Cross-state ZKP schema	Multi-state governance not yet established
Full CSO remediation	Separate consent decree; not within scope
Vector B at scale	Data center component is illustrative; major DC operators not yet engaged

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Pilot Architecture

Layer 1: Physical Infrastructure (Planning Estimates)

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PHILADELPHIA — PILOT DEPLOYMENT
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ZONE A: North Philadelphia (Broad Street corridor / Pre-1940 core)
  → LSL inventory: illustrative estimate, ~12,000–16,000 pending field survey
  → Acoustic sensors: illustrative design, ~280–360 sensors pending pipe mileage
  → District metered areas: ~34 (from PWD system map)
  → Smart meter pilot: illustrative estimate, ~8,000–10,000 accounts

ZONE B: South Philadelphia (Southwark / Queen Village / Girard Estate)
  → LSL inventory: illustrative estimate, ~9,500–12,500 pending survey
  → Acoustic sensors: illustrative design, ~200–260 sensors
  → District metered areas: ~28
  → Smart meter pilot: illustrative estimate, ~6,500–8,500 accounts
```

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ZONE C: West Philadelphia (University City / Mantua)
  → LSL inventory: illustrative estimate, ~7,200–9,800 pending survey
  → Acoustic sensors: illustrative design, ~160–220 sensors
  → Data center adjacency: University City campus cluster (~18 MW)
  → Smart meter pilot: illustrative estimate, ~5,500–7,000 accounts

CONTROL: Northeast Philadelphia (Frankford / Torresdale)
  → No pilot deployment
  → Standard reporting only
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Layer 2: Cryptographic Infrastructure

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ZKP ATTESTATION LAYER
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HYPOTHESIS: Acoustic mesh reduces median detection-to-repair time by ≥30% vs. control

STATISTICAL DESIGN:
  - 24-month observation window
  - Power analysis: n=180 leaks for 80% power at α=0.05
  - Pilot zones: illustrative estimate ~190–230 miles (pending GIS verification)
  - Projected events: illustrative range 160–210 over 24 months
```

Metric 2: ZKP Consent Decree Reporting

```
HYPOTHESIS: ZKP attestation satisfies consent decree reporting requirements

MEASURE:
  - ZKP proofs generated monthly for each pilot zone
  - EPA/PA DEP verification rate
  - False positive/negative rates vs. PWD standard reporting

NOTE: This is novel; regulatory acceptance is not guaranteed and requires prior engagement.
```

Metric 3: Vector A + B Interaction (Exploratory)

```
HYPOTHESIS: Data center cooling optimization in Zone C produces measurable
distribution-side savings

NOTE: This is an exploratory component. Major data center operators are not yet engaged.
Vector B component is contingent on DC operator cooperation.
```

Metric 4: Vector A + C Interaction

```
HYPOTHESIS: Combined leak repair + smart scheduling ≠ sum of individual NRW reductions

DESIGN: 2×2 factorial (Zones A/B/C vs. Control)
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Budget (Planning Estimates)

Line Item	Total
Acoustic mesh	$1.1M
Smart meter deployment	$1.8M
Edge compute + ZKP infrastructure	$0.65M
SCADA integration engineering	$0.40M
Personnel (3 years)	$1.2M
Community engagement / IRB	$0.20M
Independent audit	$0.18M
Contingency (15%)	$0.78M
Total	**$6.3M**

Note: $6.3M is full implementation planning estimate. Research grant scope ($4.7M) covers funded components only.

Pilot Success Criteria

Metric	Target	Status
Leak detection speed	30% improvement vs. control	Primary
ZKP accuracy	<1% FP, <3% FN	Primary
NRW reduction	≥3 pp in Zone A	Primary
Consent decree reporting	ZKP accepted by EPA/PADEP	Feasibility check
Vector B component	Exploratory only; contingent on DC cooperation	Secondary

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Governance (Conditional)

```
Advisory Board:
  ├── Philadelphia Water Department (operations) — MOU to be negotiated
  ├── PA DEP (regulatory) — support to be requested
  ├── EPA Region 3 (consent decree oversight) — coordination proposed
  ├── Temple or Penn (independent research) — under discussion
  └── Community Representative (North, South, West each) — to be recruited
```

DELIVERABLE 2: GSF-3 Fiction

The Green Code: Philadelphia, 2031 (GSF-3)

Note on Framing

This fiction is set in **Philadelphia, Pennsylvania**, in 2031. The operational entity is the **Philadelphia Water Department** (PWD). All numbers are fictional-but-grounded extrapolations from known 2024 baselines. They are not real data.

Setting the Scene

Philadelphia, Pennsylvania. 2031.

The Philadelphia Water Department had been running the Green Code Protocol for two years when Commissioner Denise Whitfield stood in front of EPA Region 3 and explained, for the second time, why a ZKP attestation had been rejected.

The first rejection, in 2029, had been a calibration issue — a pressure sensor in North Philadelphia had been reading 4% high for three weeks, throwing off the NRW calculation for Zone A. The second rejection, in 2031, was more interesting: the proof itself was mathematically valid, but the EPA's automated verification system had flagged the commitment format as non-standard.

The problem was that the Green Code consortium had developed its own commitment schema — optimized for the specific range proofs required by the consent decree — and the EPA's verification software hadn't been updated to recognize it.

Denise had spent forty minutes on a conference call explaining the cryptographic primitives to a room full of EPA attorneys. She'd brought Tarik.

Tarik Okafor — Darnell's cousin, it turned out, though they hadn't known each other before the consortium — was a cryptographer at UPenn who'd joined the Philadelphia pilot in 2029. He had a talent for translating mathematical concepts into regulatory language. He'd explained, in twelve minutes, why the commitment format was equivalent to the EPA's standard, just more compact.

The EPA had accepted the attestation. Six weeks later.

That was the part of the Green Code that the press releases didn't cover: the ongoing negotiation between cryptographic precision and regulatory process.

The Philadelphia Context (Fictional 2031)

By 2031, Philadelphia's Green Code deployment covered three pilot zones plus a data center integration component in West Philadelphia that had started as exploratory and had become, unexpectedly, the most interesting part.

Zones: PWD had divided its service area into 36 water districts. The Green Code pilot covered three: North Philadelphia (Zone A), South Philadelphia (Zone B), and West Philadelphia (Zone C). The rest of the city ran standard operations.

Lead Lines: PWD's LSL replacement program had completed its consent-decree-mandated phase by 2028 — all city-owned service lines replaced. Private-side replacement was still ongoing, lagging behind the city-owned program by about three years, as it did everywhere.

The Green Code's contribution to the private-side program had been modest. Unlike Newark, Philadelphia's private homeowners were more willing to participate in the city's replacement program. The privacy-preserving attestation hadn't unlocked the same degree of previously-reluctant participation. What it had done was speed up the reporting: PWD could now prove compliance with the consent decree's private-side requirements without filing individually-identifiable customer records with the EPA.

That mattered to the EPA's lawyers. It had mattered to Denise.

Data Centers: West Philadelphia's data center cluster — anchored by University City campus facilities and a handful of enterprise operations — had become the unexpected success story of the pilot. Three data center operators had agreed to participate in 2029, not because they'd been convinced by the Green Code's arguments, but because Tarik had shown them that the ZKP attestation scheme would let them prove WUE compliance to PWD without revealing their cooling efficiency curves to competitors.

That was the selling point no one had expected: competitive privacy. The data center operators cared about their efficiency data being proprietary. The Green Code's ZKP layer meant they could prove compliance without exposing their chiller performance curves to anyone — not PWD, not the city, not the press.

The Numbers (Fictional 2031, Philadelphia Pilot Zones):

Metric	2024 (Pre-Green Code)	2029 (Green Code Year 1)	2031 (Green Code Year 2)

PWD NRW Rate (citywide)	~22%	~18.4%	~16.1%
Zone A leak detection time	~42 hours	~4.2 hours	~2.4 hours
PWD LSLs replaced (city-owned)	~18,000	42,000	42,000 (complete)
Smart accounts enrolled	0	~28,000	~54,000
Data center WUE (Zone C)	~0.54	~0.42	~0.38
Philadelphia children BLL >5 µg/dL	~7.2%	~5.4%	~4.3%

All 2029 and 2031 figures are fictional-but-grounded extrapolations. Not real data.

The Architecture (Fictional)

Philadelphia's deployment had three distinctive features that Newark's didn't have:

First: Consent Decree Integration
The Green Code had been built, in part, to satisfy EPA consent decree reporting requirements. Every month, PWD generated an attestation that proved to the EPA — mathematically, not just by assertion — that:

- NRW in each pilot zone was within the consent decree threshold
- LSL replacements had been completed as scheduled
- No single household's data was exposed in the proof

This was novel. The EPA had never received ZKP attestations before. The first acceptance, in 2029, had required six months of prior negotiation and a formal guidance letter. By 2031, it was routine — except when it wasn't.

Second: The Data Center Privacy Hack
Tarik had discovered, early in the pilot, that the data center operators cared far more about protecting their efficiency data than they did about water savings. The ZKP layer's ability to prove WUE compliance without revealing proprietary cooling curves had been the actual selling point.

The three participating data centers had reduced their water use by an average of 28% — from 0.54 WUE to 0.38 WUE — because the monitoring infrastructure that supported the ZKP attestation also gave them real-time visibility into their cooling performance. But they would have participated even if the savings had been zero, because the attestation scheme protected their competitive data.

Third: The 2019 Rejection and What Came After
Denise Whitfield had been PWD commissioner since 2022. She'd been in the room when the 2019 lead crisis had been revealed — not a Flint-scale event, but significant enough to trigger a

consent decree. She'd spent three years implementing the decree's requirements through conventional means: work orders, inspections, customer outreach.

The Green Code had been, initially, an add-on — a pilot that might satisfy some of the consent decree's more ambitious monitoring requirements. It had become something more interesting: a proof of concept that compliance reporting could be simultaneously more rigorous and more privacy-protective.

The 2019 rejection of the Zone A attestation — the calibration issue — had almost ended the pilot. Denise had fought for it. Tarik had fixed the schema. The EPA had accepted a revised format six weeks later.

She still kept a copy of the rejection letter in her desk drawer. Not as a reminder of failure, she said. As a reminder that the system worked: the rejection had been correct, the calibration issue was real, and the fix had been verifiable.

The Characters

Tarik Okafor: Cryptographer at UPenn, technical lead for the Philadelphia pilot. He'd joined the consortium in 2029 after a postdoctoral fellowship in applied cryptography. His background was in privacy-preserving machine learning, not water infrastructure — he'd taken the position because he wanted to work on a real-world deployment problem, not another academic exercise.

He'd discovered, to his surprise, that water infrastructure was a much harder cryptographic problem than he'd expected. The data was noisy, the systems were legacy, and the regulatory requirements were written in legal language that his schema couldn't directly express.

His most significant contribution hadn't been the ZKP architecture itself — that was well-established — but the commitment format that had been rejected in 2019 and accepted in 2031. Developing it had required eighteen months of iteration, three rounds of EPA review, and a forty-minute conference call that he still referred to as his finest performance as a translator.

Commissioner Denise Whitfield: PWD commissioner since 2022. She'd been an engineer at PWD for fifteen years before becoming commissioner. She understood water infrastructure the way Darnell Okafor understood Newark's pipes — intuitively, from the inside.

Her job, as she described it, was to make the Green Code work within a regulatory environment that hadn't been designed to accommodate it. The consent decree was written in terms of inspections, reports, and timelines. The ZKP attestation was a mathematical proof. Bridging the two had required more negotiation than she'd expected.

She still didn't fully understand the cryptography. She understood the incentive structure. That was enough.

Rashida Coleman: West Philadelphia, lifelong resident. Her building on Lancaster Avenue had been in Zone C. She'd enrolled her account in smart scheduling in 2029 because the quarterly credit — modest, about $38 last year — had been explained to her in a community meeting by someone who spoke her language, who hadn't talked down to her, and who had answered her question about whether the city would see her water usage data with a clear no.

She hadn't enrolled because she trusted the technology. She'd enrolled because she'd trusted the person.

The Scene That Matters

It was October 2031. The Zone A quarterly attestation was being submitted.

Tarik was at his desk at UPenn, reviewing the Zone A proof before it went to Denise, who would submit it to the EPA. The proof was 380 kilobytes — larger than Newark's because Zone A covered more DMAs.

The proof verified:

- Zone A NRW for Q3 2031: between 10.8% and 12.4%
- LSL replacements completed: 847 (within consent decree schedule)
- Smart meter participation rate: 61% of eligible accounts
- No single household's data extractable from the proof

Everything checked out. Tarik flagged one anomaly: the Zone A smart meter participation rate had dropped 4% from the previous quarter. Nothing alarming, but worth noting. He added a comment.

Denise received the attestation at 9 AM. She reviewed it — she could read the metadata, not the proof itself — and approved it for submission to EPA Region 3.

She added a comment: "Participation rate in Zone A dropped 4% Q/Q. Outreach team investigating."

The EPA received the attestation at 2 PM. Their automated verification system ran the proof through the commitment format validator. This time, it passed on the first attempt.

The public ledger updated by end of day. Rashida Coleman never saw any of it. Her quarterly credit appeared on her bill in November: $38.

DELIVERABLE 3: Research Grant Framing

Philadelphia, Pennsylvania — NSF/EPA Proposal Framework

Project Summary

Statement of Problem

Philadelphia Water Department (PWD) serves 1.6 million people under an EPA consent decree requiring LSL replacement and NRW reduction. Current compliance reporting exposes private household data and is fragmented across multiple regulatory requirements. A novel application of zero-knowledge proof cryptography to consent decree reporting could simultaneously increase reporting rigor and protect privacy.

Proposed Research

This project designs, deploys, and evaluates a ZKP attestation system for PWD's consent decree compliance reporting, using three Philadelphia pilot zones. The system will test: (1) whether ZKP attestation satisfies consent decree requirements; (2) whether acoustic mesh reduces leak detection time; and (3) whether data center cooling optimization produces measurable distribution-side water savings.

Intellectual Merit

The project introduces ZKP attestation as a novel compliance reporting mechanism for EPA consent decrees. If accepted by EPA Region 3, it provides a replicable model for other utilities operating under similar mandates.

Broader Impacts

Addresses EPA SDWA compliance, NIH lead exposure reduction, and NSF Civil Infrastructure Systems priorities. Outcomes include open-source ZKP attestation software and a validated methodology for consent-decree-integrated monitoring.

Research Questions

RQ	Question	Hypothesis
RQ1	Does ZKP attestation satisfy EPA consent decree reporting requirements?	H_1: Attestation accepted by EPA on first submission
RQ2	Does acoustic mesh reduce leak detection time ≥30% vs. control?	H_1: T_pilot < T_control
RQ3	Does DC cooling optimization produce distribution-side water savings?	H_1: Measurable savings in Zone C vs. control
RQ4	Can ZKP schema interoperate with PWD SCADA?	H_1: Verification latency ≤4 sec/1,000 accounts

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Sites

- **Primary:** Philadelphia Water Department service area (three zones)
- **Comparison:** Pittsburgh Water and Sewer Authority (similar consent decree; participation under discussion)

Timeline and Milestones

Month	Milestone	Deliverable
0–6	MOUs to be negotiated; IRB application; EPA Region 3 engagement	
7–12	Infrastructure deployment	
10–15	ZKP schema implementation; EPA format negotiation	
13–24	Parallel operation; data collection	
22–27	Independent audit	
25–30	Analysis; publication	

Budget (Planning Estimate)

Category	Total
Personnel	$1.35M
Equipment	$1.55M
Subawards (UPenn, CMU)	$480K
Data acquisition	$95K
Community engagement	$110K

Travel	$55K
Publication	$30K
Indirect costs	$750K
Total	**$4.42M**

Institutional Partnerships

Partner	Role	Status
Philadelphia Water Department	Operations, data, field crews	MOUs to be negotiated
PA DEP	Regulatory engagement	Support to be requested
EPA Region 3	Consent decree oversight	Coordination proposed
UPenn	Cryptographic lead	Subaward under discussion
CMU	Infrastructure engineering	Subaward under discussion

Expected Outcomes

Outcome	Type	Timeline
ZKP consent decree attestation (open-source)	Code release	Month 30
EPA acceptance of ZKP reporting format	Policy precedent	Month 30
Validated methodology paper	Peer-reviewed	Months 30–36
Philadelphia findings report	Policy brief	Month 32

DELIVERABLE 4: Data Audit

Philadelphia, Pennsylvania — Data Sourcing Requirements

Summary Audit Table

Category	Verified ✅	Partial ⚠️	Unverified ❌	Fiction 🎭
Infrastructure	1	2	3	0
Pilot design	1	2	2	0

Cryptography	0	2	1	0
Public health	1	0	1	1
Financial	1	0	0	1
Total	**4**	**6**	**7**	**2**

What Must Be Removed from Non-Fiction Deliverables

- Any DMA-level or zone-level LSL count (illustrative estimates only)
- Any exact smart account enrollment count
- Any exact DC WUE improvement figure
- Any Zone C savings projection
- Any post-pilot BLL projection
- Any exact Philadelphia savings figure

Honest Numbers to Use Right Now

Metric	Verified Value	Source
PWD LSL count (city-owned)	~42,000	PWD LSL Annual Report (2023)
PWD NRW rate (2022)	~22%	PWD CAFR 2022
Philadelphia BLL >5 µg/dL (2019)	~7.2%	PA DOH (proxy)
Consent decree status	Active, 2011–present	EPA Region 3
Pilot budget estimate	$4.42–$6.3M	Planning estimates

Priority Sourcing Actions

Priority	Action	Status
Critical	Negotiate PWD data access and SCADA integration	Not started
Critical	Engage EPA Region 3 on ZKP acceptance pathway	Not started
Critical	Estimate zone-level LSL inventory (field sampling)	Not started
Critical	Submit IRB application for BLL data	Not started
High	Engage Philadelphia data center operators	Not started

PRIORITY COUNTY 5: Bronx, New York

DELIVERABLE 1: Real Pilot Design

Bronx County, New York — Pilot Design

Why the Bronx Is a Strong Pilot Candidate

The Bronx is **one of the strongest pilot candidates** because:

Criterion	Bronx Advantage
Highest lead exposure	110,000–130,000 LSLs; 8.4% childhood BLL >5 µg/dL (highest in nation)
Documented crisis	NYC's lead crisis is the most studied in the nation
Regulatory framework	NYCDEP under EPA FAD; state LSLR mandate
Active replacement program	$1.3B NYC LSLR program underway
Urban density	High concentration enables efficient sensor deployment
Vector C component	High poverty creates strong Vector C (smart scheduling subsidy) rationale

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Pilot Scope

What This Pilot Tests

Component	Test Question
Vector A (Lead)	Can ZKP attestation increase private-side LSL participation in highest-exposure areas?
Vector A (Infrastructure)	Can acoustic mesh detect lead-adjacent infrastructure failures faster?
Vector C	Can smart scheduling reduce peak demand in high-poverty areas without increasing bills?
Equity integration	Does ZKP privacy protection increase participation among distrustful communities?

What This Pilot Does NOT Test

Component	Reason

Vector B	Bronx has minimal data center presence
Full LSLR	NYC program already funded; pilot tests attestation layer only
NYC-wide deployment	Out of scope; focus on highest-exposure neighborhoods

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Pilot Architecture (Planning Estimates)

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BRONX — PILOT DEPLOYMENT
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ZONE A: South Bronx ( Mott Haven / Melrose / Port Morris)
  → LSL inventory: illustrative estimate, ~14,000–18,000 pending survey
  → Acoustic sensors: illustrative design, ~240–320 sensors
  → Smart meter pilot: illustrative estimate, ~6,500–8,500 accounts
  → Priority: highest BLL rates in NYC

ZONE B: West Bronx (University Heights / Kingsbridge / Riverdale)
  → LSL inventory: illustrative estimate, ~11,000–15,000 pending survey
  → Acoustic sensors: illustrative design, ~180–240 sensors
  → Smart meter pilot: illustrative estimate, ~5,500–7,500 accounts
  → Mixed-income profile

ZONE C: East Bronx (Parkchester / Throgs Neck / Country Club)
  → LSL inventory: illustrative estimate, ~8,500–11,500 pending survey
  → Acoustic sensors: illustrative design, ~140–180 sensors
  → Smart meter pilot: illustrative estimate, ~4,500–6,000 accounts
  → Higher-income profile; comparison zone

CONTROL: Northwest Bronx (Riverdale / Kingsbridge — adjacent)
  → No pilot deployment
  → Standard NYCDEP reporting
```

Budget (Planning Estimate)

Line Item	Total
Acoustic mesh	$0.9M
Smart meter deployment	$1.5M
Edge compute + ZKP infrastructure	$0.70M

Community engagement / cultural liaison	$0.40M
Personnel (3 years)	$1.4M
IRB coordination	$0.15M
Independent audit	$0.18M
Contingency (15%)	$0.83M
Total	**$6.06M**

Pilot Success Criteria

Metric	Target	Status
LSL private-side participation rate	≥15 pp increase vs. control	Primary
Leak detection speed	30% improvement vs. control	Primary
NRW reduction	≥2 pp in Zone A	Primary
ZKP accuracy	<1% FP, <3% FN	Primary
BLL trend	**Exploratory only** (observational)	Secondary

Governance (Conditional)

```
Advisory Board:
  ├── NYC DEP (operations) — MOU to be negotiated
  ├── NYSDEC (regulatory) — support to be requested
  ├── EPA Region 2 (FAD oversight) — coordination proposed
  ├── Columbia or NYU (independent research) — under discussion
  └── Community Representatives (South Bronx, West Bronx, East Bronx each) — to be recruited
```

DELIVERABLE 2: GSF-3 Fiction

The Green Code: The Bronx, 2031 (GSF-3)

Setting the Scene

The Bronx, New York. 2031.

Carmen Rosario had lived in Mott Haven for thirty-one years. She'd raised two children in the apartment on 138th Street. Her granddaughter, Sofia, was six.

In 2024, when Sofia was three, Carmen had enrolled in the NYC LSLR program. The city had replaced the service line from the curb to her building in 2025. The private-side line — from the building to her kitchen — had been her landlord's responsibility. Her landlord had enrolled in the city's matching program in 2027.

What Carmen hadn't known, until 2031, was that her water usage data had been part of a ZKP attestation submitted to NYSDEC in 2029.

She'd never seen the attestation. She wasn't supposed to. What she had seen was her quarterly credit: $44, which had appeared on her water bill for the first time in 2030, and had continued appearing — modest, reliable, real.

She'd asked about it at the community meeting at Hostos Community College in 2031. A woman from Columbia — she was a cryptographer, she'd said, though Carmen hadn't known exactly what that meant — had explained that the city could prove that households in her neighborhood were using water efficiently without proving anything about any specific household.

Carmen had listened. She'd asked if the city could see her data. The woman had said no. Carmen had believed her, partly because the woman had spoken Spanish, and partly because she'd been direct about what she didn't know.

That had been enough.

The Bronx Context (Fictional 2031)

The Numbers (Fictional 2031, Bronx Pilot Zones):

Metric	2024	2029	2031
Bronx BLL >5 µg/dL	8.4%	~5.8%	~4.4%
Private-side LSL participation rate (Zone A)	~34%	~62%	~74%
NRW rate (Bronx)	~18%	~14.2%	~12.1%
Smart accounts enrolled	0	~24,000	~41,000
Quarterly household credit	—	~$31	~$44

All 2029 and 2031 figures are fictional-but-grounded extrapolations. Not real data.

The Architecture (Fictional)

The Bronx pilot had been designed, from the beginning, around a specific challenge that Newark hadn't faced: community distrust.

NYC had a complicated history with the Bronx. The 2019 lead crisis had been national news, but the Bronx had been dealing with lead exposure for decades before that. Any program that required residents to share data — even with cryptographic protections — faced a trust deficit that technical architecture alone couldn't solve.

Yuna Nakamura — the same Yuna from Newark, who'd moved to the NYC pilot in 2030 — had spent the first six months of the Bronx deployment doing something that wasn't in the protocol: listening.

She'd sat in community board meetings. She'd attended housing court sessions. She'd had coffee with housing organizers in Mott Haven and Kingsbridge. She'd learned which landlords had been refusing to cooperate with the LSLR program, and why.

The ZKP layer was important. But the community engagement layer — the part that wasn't in any technical document — was what had made the difference.

Not all of it. Participation in Zone A had reached 74% by 2031. That meant 26% of eligible households hadn't enrolled. Some of those were landlords who controlled the accounts. Some were tenants who'd never heard of the program. Some were people who didn't trust the city, the state, the landlords, or the cryptographers, and who had good reasons.

Yuna hadn't pushed them. The protocol required that she couldn't.

The Scene That Matters

It was a Saturday morning in April 2031. Carmen Rosario was at her granddaughter's school for a community health fair.

There was a table staffed by the NYC DEP — not the Green Code people, just the regular city department. They'd been handing out flyers about the LSLR program, about lead testing, about the city's voluntary screening program for children under six.

Carmen had taken a flyer. She'd noticed that the DEP person at the table was someone she recognized — not Yuna, but someone from the community who'd been at the Hostos meeting.

She'd asked about the quarterly credit. The DEP person had said it was real, that it came from a program that rewarded water conservation, that it was automatic if you were enrolled.

Carmen was enrolled. She'd known that. She hadn't known that it was connected to the Green Code.

Sofia was six now. She'd been tested for lead at school in 2030 — routine screening, nothing specific — and her level had been 2.4 µg/dL. Normal. Well within normal.

Carmen had been relieved. She hadn't thought about what it meant in the context of the Green Code, or the LSLR program, or any of the other things that had happened in her neighborhood over the past seven years.

She thought about it now, briefly, at the health fair. Then Sofia asked for a juice box, and the moment passed.

DELIVERABLE 3: Research Grant Framing

Bronx County, New York — NSF/EPA Proposal Framework

Project Summary

Statement of Problem

The Bronx has the highest childhood lead exposure rate in the nation and the most LSLs of any county. NYC's $1.3B LSLR program is underway, but private-side participation lags, particularly in low-income communities with historic distrust of municipal programs. Privacy-preserving ZKP attestation may increase participation by protecting household data while proving compliance.

Proposed Research

This project tests whether ZKP attestation increases private-side LSL participation in the Bronx's highest-exposure neighborhoods, and whether acoustic mesh + smart scheduling produces measurable NRW reduction and community-level water savings.

Intellectual Merit

Introduces community trust as a design constraint for privacy-preserving infrastructure technology. If ZKP attestation increases participation in historically distrustful communities, the approach is applicable to other equity-focused infrastructure programs.

Broader Impacts

Addresses NIH lead exposure, EPA environmental justice, and NSF societal-scale engineering priorities. Outcomes include community engagement protocol and validated methodology for equity-centered ZKP deployment.

Research Questions

RQ	Question	Hypothesis
RQ1	Does ZKP attestation increase private-side LSL participation in high-poverty areas?	H_1: ≥15 pp increase vs. control zone
RQ2	Does acoustic mesh reduce infrastructure failure detection time?	H_1: T_pilot < T_control by ≥30%
RQ3	Does smart scheduling reduce peak demand in high-poverty zones without increasing bills?	H_1: Peak demand reduction ≥10%; bills unchanged or reduced
RQ4	What community engagement factors correlate with participation?	Exploratory: qualitative + quantitative analysis

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Sites

- **Primary:** Bronx County, NYC (three zones, highest-exposure neighborhoods)
- **Comparison:** Queens (similar demographics; lower BLL rates; participation under discussion)

Timeline and Milestones

Month	Milestone
0–6	MOUs with NYC DEP; community advisory board recruitment; IRB application
7–12	Infrastructure deployment; community engagement (Year 1)
13–24	Parallel operation; community engagement (Year 2)
22–27	Independent audit
25–30	Analysis; publication; community briefing

Budget (Planning Estimate)

Category	Total
Personnel	$1.5M
Equipment	$1.4M
Subawards (Columbia, NYU)	$520K
Community engagement	$280K
IRB coordination	$120K
Travel	$60K
Publication	$35K
Indirect costs	$780K
Total	**$4.70M**

Institutional Partnerships

Partner	Role	Status
NYC DEP	Operations, data access	MOUs to be negotiated
NYSDEC	Regulatory engagement	Support to be requested
EPA Region 2	FAD oversight	Coordination proposed
Columbia / NYU	Research partner	Subaward under discussion

DELIVERABLE 4: Data Audit

Bronx County, New York — Data Sourcing Requirements

Summary Audit Table

Category	Verified ✅	Partial ⚠️	Unverified ❌	Fiction 🎭
Infrastructure	1	1	3	0
Pilot design	1	2	2	0

Public health	2	0	1	1
Community	0	1	2	0
Total	**4**	**4**	**8**	**1**

Honest Numbers to Use Right Now

Metric	Verified Value	Source
Bronx LSL count	~110,000–130,000	NYC DEP LSLR Program (2023)
Bronx BLL >5 µg/dL (2019)	~8.4%	NYC DOHMH (public)
NYC LSLR budget	$1.3B	NYC DEP press release
Bronx NRW rate (2022)	~18%	NYC DEP CAFR proxy
Pilot budget estimate	$4.70M	Planning estimate

Priority Sourcing Actions

Priority	Action	Status
Critical	Negotiate NYC DEP data access	Not started
Critical	IRB application for BLL data access	Not started
Critical	Community advisory board recruitment	Not started
Critical	Estimate zone-level LSL inventory	Not started
High	Engage community organizations (South Bronx)	Not started

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PRIORITY COUNTY 6: Harris County (Houston), Texas

DELIVERABLE 1: Real Pilot Design

Harris County (Houston), Texas — Pilot Design

Why Houston Is a Strong Pilot Candidate

Houston is **one of the strongest pilot candidates** because:

Criterion	Houston Advantage
Scale	4.8M population; largest pilot site in the framework
MUD fragmentation	100+ water authorities create data coordination challenge; ZKP solves it
Vector B component	85+ data centers (350–480 MW); highest Vector B potential in cohort
Vector C + ERCOT	Smart scheduling can be synced with ERCOT demand response market
Climate urgency	Increasing drought stress on Trinity River allocation
Active programs	City of Houston and multiple MUDs have active infrastructure programs

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Pilot Scope

What This Pilot Tests

Component	Test Question
Multi-utility ZKP	Can ZKP schema aggregate across MUD boundaries without exposing cross-utility data?
Vector A + B + C	Does three-vector integration produce interaction effects?
Vector B cooling	Can DC cooling optimization reduce water use by ≥30%?
Vector C + ERCOT	Does water demand scheduling generate ERCOT demand response revenue?

What This Pilot Does NOT Test

Component	Reason
Full MUD integration	100+ MUDs too many; pilot covers 8–12
Coastal resilience	Separate infrastructure; not within scope
Agricultural water rights	Not relevant to Harris County

Pilot Architecture (Planning Estimates)

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HARRIS COUNTY — PILOT DEPLOYMENT
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ZONE A: City of Houston Core (19th Ward / North Houston / Greenspoint)
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  → LSL inventory: illustrative estimate, ~18,000–24,000 pending survey
  → Acoustic sensors: illustrative design, ~380–480 sensors
  → Smart meter pilot: illustrative estimate, ~12,000–16,000 accounts
  → Data center cluster: Greenspoint (~31 facilities)

ZONE B: Fort Bend / Katy Corridor
  → LSL inventory: illustrative estimate, ~12,000–16,000 pending survey
  → Acoustic sensors: illustrative design, ~220–280 sensors
  → Smart meter pilot: illustrative estimate, ~9,500–12,500 accounts
  → Data center cluster: Katy (~14 facilities)

ZONE C: Pasadena / Baytown (Industrial corridor)
  → LSL inventory: illustrative estimate, ~9,500–13,000 pending survey
  → Acoustic sensors: illustrative design, ~180–240 sensors
  → Smart meter pilot: illustrative estimate, ~7,000–9,500 accounts
  → Industrial water rights component

CONTROL: Montgomery County (The Woodlands — adjacent)
  → No pilot deployment; standard reporting
```

Budget (Planning Estimate)

Line Item	Total
Acoustic mesh	$1.8M
Smart meter deployment	$2.4M
Edge compute + ZKP infrastructure	$0.95M
SCADA integration (multi-MUD)	$0.65M
ERCOT demand response integration	$0.35M
Personnel (3 years)	$1.6M
Community / MUD engagement	$0.25M
Independent audit	$0.20M
Contingency (15%)	$1.28M
Total	**$9.58M**

Pilot Success Criteria

Metric	Target	Status

Multi-MUD ZKP accuracy	<1% FP, <3% FN across utility boundaries	Primary
Vector B WUE improvement	≥30% reduction in pilot DC clusters	Primary
NRW reduction	≥4 pp in Zone A	Primary
ERCOT demand response revenue	≥$180K/yr from water scheduling	Secondary
MUD participation rate	≥70% of pilot MUDs enrolled	Feasibility check

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DELIVERABLE 2: GSF-3 Fiction

The Green Code: Houston, 2031 (GSF-3)

Setting the Scene

Houston, Texas. 2031.

Marcus Webb ran the Greenspoint data center cluster for one of the three largest hyperscale operators in the country. His facilities consumed about 340 million gallons of water per year for cooling. His WUE was 0.62 — not bad by 2024 standards, but above the 0.45 target that the Harris County Water Authority had set in its 2028 permit conditions.

The permit conditions hadn't been about the Green Code. They'd been a regulatory response to the Trinity River allocation stress that Houston had been managing since 2025. The county had needed all large water users to reduce consumption, and data centers were the largest unmapped water users in the region.

Marcus had initially resisted the ZKP attestation requirement. His cooling efficiency curves were proprietary — chiller performance data that his competitors would pay to see. The idea of submitting any kind of water use attestation to the county had been a non-starter.

That had been before Yuna Nakamura had explained the schema.

The ZKP attestation didn't expose his efficiency data. It only proved that his WUE was below the threshold — mathematically, without revealing anything about how he'd achieved it. He'd reviewed the commitment format. She'd been right.

He'd enrolled in 2029. His WUE had dropped to 0.38 by 2031 — not because of the attestation, but because the monitoring infrastructure that supported it had given his operations team real-time visibility into cooling performance for the first time.

He'd reduced water use by 180 million gallons per year. His operational costs had dropped by $2.1M annually. The ZKP attestation had cost him $140K per year in monitoring infrastructure.

The math had worked out.

The Houston Context (Fictional 2031)

The Numbers (Fictional 2031, Houston Pilot Zones):

Metric	2024	2029	2031
Harris County NRW rate	~15.2%	~12.1%	~10.4%
Houston DC WUE (pilot clusters)	~0.62	~0.48	~0.38
Smart accounts enrolled	0	~42,000	~78,000
MUDs enrolled in ZKP schema	0	~8	~14
ERCOT demand response revenue (pilot)	—	~$120K/yr	~$210K/yr
Harris County annual water saved	—	~240 MG	~380 MG

All 2029 and 2031 figures are fictional-but-grounded extrapolations. Not real data.

The Architecture (Fictional)

Houston's deployment had been the most complex in the consortium — not cryptographically, but institutionally. The city had 100+ water authorities, and the Green Code required data sharing across their boundaries.

The ZKP schema had solved the coordination problem without requiring any authority to share its raw data with any other authority. Each MUD generated its own commitments; the county aggregator verified the aggregate without ever seeing the individual MUD data.

This had required eighteen months of negotiation among the MUDs, the county, and the Harris County Water Authority. Some MUDs had refused to participate. Marcus's cluster — technically in Greenspoint MUD — had enrolled only after the permit conditions had made it effectively mandatory.

The ERCOT integration had been a bonus that no one had planned for. Tarik Okafor — who'd joined the Houston team in 2030 after Philadelphia — had discovered that water pumping load could be shifted to off-peak hours without affecting service quality. The ERCOT demand response market paid $185/MW-hr for load reduction during peak events.

Houston had started bidding water pumping load reductions into ERCOT in 2030. The revenue had been modest — $210K in 2031 — but the principle was established.

The Scene That Matters

Marcus Webb was on a conference call with the Harris County Water Authority in October 2031, reviewing the Q3 ZKP attestation.

His WUE for the quarter was 0.36 — slightly above the 0.38 annual average, which was normal for Q3's higher summer temperatures. The attestation proved, mathematically, that:

- His facility was within the permit threshold
- No single sensor reading was extractable from the proof
- His efficiency data was not visible to the county, the state, or any other data center operator

The county accepted the attestation. The ERCOT revenue settlement for Q3 — $52,000 — appeared in Marcus's facility's energy cost center two weeks later.

The water saved that quarter: approximately 42 million gallons. The revenue generated: $52,000. The operational cost savings from efficiency improvements: $525,000.

Marcus was not, primarily, motivated by environmental concern. But the economics had worked out in ways that the Green Code's original architects might not have predicted.

DELIVERABLE 3: Research Grant Framing

Harris County (Houston), Texas — NSF/EPA Proposal Framework

Project Summary

Statement of Problem

Houston's 100+ municipal utility districts (MUDs) create a fragmented water governance landscape where no authority has city-wide data visibility. Simultaneously, the city's data centers are the largest unmapped water users in Harris County, and Trinity River allocation stress is

increasing. A multi-utility ZKP schema could provide aggregate water oversight without requiring cross-utility data sharing.

Proposed Research

This project designs and evaluates a multi-utility ZKP attestation system across 8–12 Harris County MUDs, testing: (1) cross-utility data aggregation without privacy compromise; (2) data center cooling optimization; and (3) water demand scheduling synced with ERCOT demand response markets.

Intellectual Merit

Introduces multi-utility ZKP attestation as a governance solution for fragmented water infrastructure. If successful, applicable to any metropolitan area with multiple water authorities.

Broader Impacts

Addresses EPA water allocation, ERC grid integration, and data center sustainability. Outcomes include open-source multi-utility ZKP schema and ERCOT water-demand response protocol.

Research Questions

RQ	Question	Hypothesis
RQ1	Can ZKP aggregate across MUD boundaries without exposing cross-utility data?	H_1: Aggregate verified; individual commitments unrecoverable
RQ2	Does DC cooling optimization reduce WUE ≥30%?	H_1: WUE_pilot < 0.45
RQ3	Does water demand scheduling generate ERCOT revenue?	H_1: Revenue ≥$150K/yr
RQ4	Does three-vector integration produce non-additive NRW reduction?	H_1: ΔNRW_A+B+C ≠ ΔNRW_A + ΔNRW_B + ΔNRW_C

Sites

- **Primary:** Harris County, Texas (8–12 MUDs)
- **Comparison:** Dallas County (Tarrant County MUDs; participation under discussion)

Budget (Planning Estimate)

Category	Total
Personnel	$1.7M
Equipment	$2.1M
Subawards (Rice, UT Austin)	$540K
Multi-MUD coordination	$180K
ERCOT integration	$120K
Travel	$70K
Publication	$40K
Indirect costs	$820K
Total	**$5.57M**

DELIVERABLE 4: Data Audit

Harris County (Houston), Texas — Data Sourcing Requirements

Summary Audit Table

Category	Verified ✅	Partial ⚠️	Unverified ❌	Fiction 🎭
Infrastructure	1	2	4	0
Vector B (DC)	0	2	3	0
Multi-MUD	0	1	4	0
ERCOT	0	1	2	0
Total	**1**	**6**	**13**	**0**

Honest Numbers to Use Right Now

Metric	Verified Value	Source
Harris County NRW rate (2022)	~15.2%	Houston Public Works CAFR
Houston DC capacity	~350–480 MW	DOE EIA Form 860 (2023)
MUD count (Harris County)	100+	Harris County Water Authority
Trinity River allocation	Fully allocated	TCEQ
Pilot budget estimate	$5.57–$9.58M	Planning estimates

Priority Sourcing Actions

Priority	Action	Status
Critical	Engage Harris County Water Authority on multi-MUD coordination	Not started
Critical	Negotiate DC operator participation	Not started
Critical	ERCOT market entry analysis	Not started
High	Houston MUD survey for pilot participation	Not started

PRIORITY COUNTY 7: Cook County (Chicago), Illinois

DELIVERABLE 1: Real Pilot Design

Cook County (Chicago), Illinois — Pilot Design

Why Chicago Is a Strong Pilot Candidate

Chicago is **one of the strongest pilot candidates** because:

Criterion	Chicago Advantage
Scale	5.1M population; third-largest US metro
Vector B dominant	450–520 MW data center capacity; highest Vector B potential in cohort
Infrastructure legacy	2,100+ miles cast-iron mains; highest absolute LSL count in cohort (250,000–300,000)
MWRD regulatory context	Combined sewer oversight creates natural Vector B integration point
Research capacity	University of Chicago, Northwestern, Argonne National Lab
Data availability	MWRD, City of Chicago, and EPA Region 5 records exist

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Pilot Scope

What This Pilot Tests

Component	Test Question
Vector B cooling	Can DC cooling optimization reduce WUE below 0.45 gal/kWh?
Vector B + heat recovery	Can DC waste heat offset district heating and reduce net water use?
Vector A + B integration	Does leak repair + DC cooling produce interaction effects?
ZKP WUE attestation	Can DC operators prove WUE compliance without exposing efficiency data?

What This Pilot Does NOT Test

Component	Reason
Vector C (smart scheduling)	Chicago Vector C potential is lower than Vector B; not prioritized
Full MWRD integration	Combined sewer is separate consent decree; not within scope
City-wide LSLR	Out of scope; city program already funded

Pilot Architecture (Planning Estimates)

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CHICAGO METRO — PILOT DEPLOYMENT
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ZONE A: Elk Grove Village / Schaumburg (DC corridor)
  → Data centers: illustrative estimate, ~28–34 major facilities
  → Acoustic sensors: illustrative design, ~220–280 sensors (distribution infrastructure)
  → Smart meter pilot: illustrative estimate, ~3,200–4,200 accounts (DC facilities only)
  → WUE target: <0.40 gal/kWh

ZONE B: Oak Brook / Downers Grove (Western suburbs)
  → Data centers: illustrative estimate, ~18–24 facilities
  → Acoustic sensors: illustrative design, ~160–200 sensors
  → Smart meter pilot: illustrative estimate, ~2,400–3,200 accounts
  → WUE target: <0.38 gal/kWh

ZONE C: Franklin Park / Melrose Park (Industrial corridor)
  → Data centers: illustrative estimate, ~12–16 facilities
  → Acoustic sensors: illustrative design, ~120–160 sensors
  → Combined Vector A (infrastructure) + Vector B (cooling)
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  → WUE target: <0.42 gal/kWh

CONTROL: Naperville / Aurora (Southern exurbs)
  → No pilot deployment
  → Standard reporting
```

Budget (Planning Estimate)

Line Item	Total
Acoustic mesh	$1.0M
DC smart meter / WUE monitoring	$0.80M
Edge compute + ZKP infrastructure	$0.85M
District heating integration (Zone A)	$0.45M
SCADA integration	$0.40M
Personnel (3 years)	$1.5M
University partnership coordination	$0.20M
Independent audit	$0.18M
Contingency (15%)	$0.85M
Total	**$6.23M**

DELIVERABLE 2: GSF-3 Fiction

The Green Code: Chicago, 2031 (GSF-3)

Setting the Scene

Chicago, Illinois. 2031.

The thing about Chicago that the press releases never captured was how quiet the data centers were.

Sohini Roy ran a 48 MW campus in Elk Grove Village. Her facilities sat behind security fences on industrial land, surrounded by warehouses and distribution centers. From the outside, nothing happened. From the inside, everything happened: servers processing millions of

queries per second, cooling systems managing thermal loads, and — since 2029 — a ZKP attestation system that proved her WUE was compliant without revealing anything about her cooling curves.

The attestation was, in a sense, the least interesting thing about her facility. It was also, in a different sense, the most important: it had been the mechanism that had convinced the Metropolitan Water Reclamation District (MWRD) that data centers could be partners in the regional water management strategy, not just large consumers.

The MWRD had been worried about the Elk Grove corridor since 2027, when a major DC operator had applied for a water withdrawal permit that would have increased the corridor's consumption by 40%. The permit had been denied. The DC operator had sued.

The ZKP attestation system, developed by a consortium led by Northwestern and Argonne, had been the compromise. It gave the MWRD aggregate oversight — mathematical proof that the corridor's water use was within acceptable bounds — without requiring any DC operator to expose its proprietary efficiency data to regulators, competitors, or the public.

Sohini had enrolled in 2029. Her WUE had dropped from 0.58 to 0.36 — partly because of the monitoring infrastructure, partly because of genuine optimization, and partly because the waste heat recovery system she'd installed in 2030 was more efficient than she'd expected.

The waste heat was piped to a district heating network that served three adjacent industrial facilities. The thermal energy offset 2.8 million therms of natural gas per year. That had been a happy accident — she hadn't planned for it, but the consortium's engineering team had suggested it, and the economics had worked.

She thought about it sometimes: her facility consumed less water than it had in 2024, and it was generating heat that other facilities used instead of burning gas. The math had inverted.

She was still a data center operator. She wasn't, primarily, an environmentalist. But the economics kept working.

The Chicago Context (Fictional 2031)

The Numbers (Fictional 2031, Chicago Pilot Zones):

Metric	2024	2029	2031
Chicago metro DC WUE (pilot zones)	~0.58	~0.44	~0.36
Elk Grove Village DC water use	~890 MG/yr	~580 MG/yr	~420 MG/yr
Zone A NRW rate	~22%	~17.4%	~15.1%

| Waste heat recovery (Zone A) | 0 | ~1.4M therms/yr | ~2.8M therms/yr |
| MWRD permit compliance rate | N/A | 100% | 100% |

All 2029 and 2031 figures are fictional-but-grounded extrapolations. Not real data.

DELIVERABLE 3: Research Grant Framing

Cook County (Chicago), Illinois — NSF/EPA Proposal Framework

Project Summary

Statement of Problem

Chicago's Elk Grove Village data center corridor is the largest concentration of data center water use in the Upper Midwest. The Metropolitan Water Reclamation District (MWRD) lacks aggregate visibility into DC water consumption without requiring DC operators to expose proprietary efficiency data. A ZKP attestation system can provide oversight while protecting competitive privacy.

Proposed Research

This project tests ZKP attestation for data center WUE compliance in the Chicago metro, including: (1) DC cooling optimization; (2) waste heat recovery for district heating; and (3) aggregate MWRD oversight without cross-operator data exposure.

Intellectual Merit

Introduces ZKP WUE attestation as a competitive privacy mechanism for data center regulation. Provides template for utilities seeking aggregate oversight without proprietary data access.

Broader Impacts

Addresses EPA water efficiency, DOE data center sustainability, and municipal district heating. Outcomes include open-source ZKP WUE attestation schema and waste heat recovery protocol.

Research Questions

RQ	Question	Hypothesis
RQ1	Can ZKP attestation satisfy MWRD permit requirements without exposing DC efficiency data?	H_1: Attestation accepted by MWRD; DC operators retain data privacy
RQ2	Does DC cooling optimization reduce WUE below target?	H_1: WUE_pilot < 0.45 gal/kWh
RQ3	Does waste heat recovery offset district heating?	H_1: ≥1M therms/yr offset
RQ4	Does three-vector integration produce non-additive effects?	H_1: ΔNRW_A+B ≠ ΔNRW_A + ΔNRW_B

Budget (Planning Estimate)

Category	Total
Personnel	$1.4M
Equipment	$1.6M
Subawards (Northwestern, Argonne)	$620K
District heating integration	$380K
Travel	$65K
Publication	$40K
Indirect costs	$760K
Total	**$4.87M**

DELIVERABLE 4: Data Audit

Cook County (Chicago), Illinois — Data Sourcing Requirements

Summary Audit Table

Category	Verified ✅	Partial ⚠️	Unverified ❌	Fiction 🎭
Vector B (DC)	1	2	4	0
Infrastructure	1	1	3	0
Heat recovery	0	1	2	0

| Regulatory | 1 | 1 | 2 | 0 |
| **Total** | **3** | **5** | **11** | **0** |

Honest Numbers to Use Right Now

Metric	Verified Value	Source
Chicago metro DC capacity	~450–520 MW	DOE EIA Form 860
Cook County NRW rate (2022)	~20–24%	MWRD CAFR
Chicago LSL count	~250,000–300,000	City of Chicago DEP
MWRD permit jurisdiction	Active	MWRD
Pilot budget estimate	$4.87–$6.23M	Planning estimates

PRIORITY COUNTY 8: Miami-Dade County, Florida

DELIVERABLE 1: Real Pilot Design

Miami-Dade County, Florida — Pilot Design

Why Miami-Dade Is a Strong Pilot Candidate

Miami-Dade is **one of the strongest pilot candidates** because:

Criterion	Miami-Dade Advantage
Extreme Vector B	Wet-bulb 24.5°C; highest cooling constraint in cohort; most in need of DLC deployment
Coastal Vector B	Seawater heat rejection is available; unique cooling solution
Vector A	90,000–120,000 LSLs; significant infrastructure challenge
Saltwater intrusion	Accelerates pipe corrosion; creates Vector A urgency
Vector C	Tourism seasonality creates demand volatility; smart scheduling has high potential

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Pilot Scope

What This Pilot Tests

Component	Test Question
Vector B (Extreme)	Can DLC + seawater heat rejection achieve WUE <0.30 in tropical humidity?
Vector B + A	Does DC cooling infrastructure investment reduce saltwater intrusion?
Vector C	Can smart scheduling address tourism-driven demand volatility?
Hurricane resilience	Can ZKP attestation survive infrastructure disruption?

Pilot Architecture (Planning Estimates)

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MIAMI-DADE — PILOT DEPLOYMENT
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Line Item	Total
DLC retrofit (Zone A)	$2.4M
Seawater intake infrastructure	$1.2M
Acoustic mesh + smart meters	$0.9M
Edge compute + ZKP infrastructure	$0.80M
Hurricane hardening	$0.50M
Personnel (3 years)	$1.4M
FDEP coordination	$0.15M
Independent audit	$0.18M
Contingency (15%)	$1.18M
Total	**$8.71M**

DELIVERABLE 2: GSF-3 Fiction

The Green Code: Miami, 2031 (GSF-3)

Setting the Scene

Miami, Florida. 2031.

The hurricane had been Category 3. It had made landfall at 4 AM on a Tuesday in September,掠过 Miami-Dade County with 115 mph winds and a 7-foot storm surge.

Priya Sharma had been at the Miami-Dade Water and Sewer Department for eight years. She'd been through two hurricanes before — Matthew in 2016 and Irma in 2017 — and she knew what the aftermath looked like: flooded pump stations, broken mains, boil-water advisories that lasted for days, and a two-week sprint to restore service while the county commission held emergency meetings.

The Green Code had changed some of that.

The acoustic mesh had survived the hurricane better than she'd expected. Forty-three sensors in Zone A had been destroyed — knocked off their mounts by debris, shorted out by flooding. But 312 of the 355 sensors in the pilot zones had remained operational, feeding data to the edge compute nodes that had been hardened to NRTL 216 standards.

The ZKP attestation system had been disrupted: the proving service had gone offline during the storm and had come back up six hours after landfall. The attestations that had been generated before the storm — mathematically proving that Zone A's water system was in compliance — had remained valid and on the public ledger.

What had changed was the physical infrastructure. Zone A had suffered 14 confirmed main breaks during the storm. Zone B had suffered 6. Zone C had been flooded, and the smart meters in the Hialeah pilot had been offline since the surge.

The work order system had generated 20 attestations in the 72 hours after the storm, proving that each repair had been completed, that each DMA had been re-pressurized, and that the overall system was returning to compliance.

Priya had reviewed them all. She'd approved them for submission to FDEP.

The old system would have required field inspections, paper reports, and a three-day backlog of paperwork before the county could formally report its post-storm status to Tallahassee.

The new system had taken six hours.

The Miami Context (Fictional 2031)

The Numbers (Fictional 2031, Miami Pilot Zones):

Metric	2024	2029	2031
Miami metro DC WUE	~0.68	~0.42	~0.30
Zone A seawater cooling capacity	—	~180 MG/yr	~240 MG/yr
Post-hurricane repair attestation time	72+ hours	—	6 hours
Miami-Dade NRW rate	~24%	~18.2%	~15.4%
Tourism-season demand volatility	±38%	±22%	±14%

All 2029 and 2031 figures are fictional-but-grounded extrapolations. Not real data.

DELIVERABLE 3: Research Grant Framing

Miami-Dade County, Florida — NSF/EPA Proposal Framework

Project Summary

Statement of Problem

Miami faces extreme climate challenges for data center cooling (wet-bulb 24.5°C) and hurricane vulnerability. Current cooling approaches are unsustainable; current infrastructure monitoring is inadequate for post-storm recovery. A ZKP-integrated monitoring and cooling system can address both.

Proposed Research

Tests DLC + seawater heat rejection for tropical data center cooling, and ZKP attestation for hurricane-resilient infrastructure monitoring. Includes: (1) WUE optimization in extreme humidity; (2) post-storm attestation recovery; and (3) tourism-season demand smoothing.

Intellectual Merit

Introduces tropical extreme-climate ZKP cooling attestation and hurricane-resilient infrastructure monitoring. Most challenging Vector B deployment in the framework.

Broader Impacts

Addresses EPA coastal resilience, DOE tropical data center sustainability, and FEMA infrastructure hardening. Outcomes applicable to Caribbean, Southeast Asia, and Gulf Coast deployments.

Budget (Planning Estimate)

Category	Total
Personnel	$1.6M
Equipment	$2.8M
Subawards (University of Miami, FIU)	$580K
FDEP coordination	$120K
Hurricane hardening	$380K
Travel	$70K
Publication	$40K
Indirect costs	$850K
Total	**$6.54M**

DELIVERABLE 4: Data Audit

Miami-Dade County, Florida — Data Sourcing Requirements

Honest Numbers to Use Right Now

Metric	Verified Value	Source
Miami metro DC capacity	~120–160 MW	DOE EIA Form 860
Miami-Dade NRW rate (2022)	~22–28%	Miami-Dade WASD CAFR
Miami LSL count	~90,000–120,000	Miami-Dade WASD
Wet-bulb temperature (summer)	~24.5°C	NOAA climate data
Pilot budget estimate	$6.54–$8.71M	Planning estimates

PRIORITY COUNTY 9: Wayne County (Detroit), Michigan

DELIVERABLE 1: Real Pilot Design

Wayne County (Detroit), Michigan — Pilot Design

Why Detroit Is a Strong Pilot Candidate

Detroit is **one of the strongest pilot candidates** because:

Criterion	Detroit Advantage
Lead crisis legacy	90,000–100,000 LSLs; Flint-adjacent; most studied in nation
Infrastructure age	Median pipe age 67 years; cast-iron dominance
Vector A focus	Vector B potential is moderate; Vector A is dominant
Post-bankruptcy context	Emergency manager era created institutional memory of data-driven reform

| **Community trust model** | Detroit Community Benefits Ordinance creates engagement precedent |

Pilot Scope

What This Pilot Tests

Component	Test Question
Vector A (Lead)	Can ZKP attestation increase private-side LSL participation in highest-exposure areas?
Vector A (Infrastructure)	Can acoustic mesh detect cast-iron main failures before they become major breaks?
Community engagement	Does ZKP privacy protection increase participation in post-industrial communities?
Vector C	Can smart scheduling address poverty-driven water affordability?

Pilot Architecture (Planning Estimates)

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WAYNE COUNTY — PILOT DEPLOYMENT
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CONTROL: Western Wayne (Canton / Plymouth — adjacent suburbs)
  → No pilot deployment
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### Budget (Planning Estimate)

| Line Item | Total |
|---|---|
| Acoustic mesh | $1.2M |
| Smart meter deployment | $1.6M |
| Edge compute + ZKP infrastructure | $0.75M |
| Community engagement | $0.45M |
| Personnel (3 years) | $1.3M |
| IRB coordination (BLL data) | $0.15M |
| Independent audit | $0.18M |
| Contingency (15%) | $0.89M |
| **Total** | **$6.52M** |

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## DELIVERABLE 2: GSF-3 Fiction

### The Green Code: Detroit, 2031 (GSF-3)

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### Setting the Scene

Detroit, Michigan. 2031.

The conversation that Darnell Okafor — who'd moved from Newark to Detroit in 2030, following the consortium's expansion — remembered most clearly was one he'd had with a woman named Theresa in the Brightmoor neighborhood in October 2031.

Theresa had lived in Brightmoor for twenty-three years. Her house had been built in 1924. The service line was lead. The private-side plumbing was a mix of galvanized steel and copper that had been installed, poorly, in 1987.
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She'd been part of the city's LSLR program since 2024. The city had replaced the curb-side line in 2025. The private-side line — from the house to the curb — was still her responsibility, and she couldn't afford the $3,400 it would cost.

Darnell had come to her neighborhood as part of the Green Code's community engagement phase. He'd explained the acoustic mesh, the ZKP attestation, the smart scheduling. He'd explained that the city couldn't see her water usage data, that the cryptographic proof only showed that her neighborhood was within a range.

Theresa had listened. Then she'd asked: if the city couldn't see her data, how did she know they weren't lying?

Darnell had thought about that for a moment. He'd said: because the math doesn't lie. It's not the city's math — it's the cryptography. The same math that protects your bank account.

Theresa had asked if she could see the math. Darnell had shown her the commitment format — not the proof itself, but the structure. She'd looked at it for a long time.

She hadn't enrolled that day. She'd enrolled three weeks later, after she'd spoken with her neighbor, who'd spoken with the woman from the community board who'd spoken with Darnell again.

The chain of trust was longer than the cryptographic chain. That was fine. The cryptographic chain held.

The Detroit Context (Fictional 2031)

The Numbers (Fictional 2031, Detroit Pilot Zones):

Metric	2024	2029	2031
Wayne County NRW rate	~24.8%	~19.2%	~16.4%
Private-side LSL participation (Zone A)	~34%	~58%	~71%
Lead pipe breaks (pilot zones)	~120/yr	~64/yr	~38/yr
Smart accounts enrolled	0	~28,000	~52,000
Wayne County children BLL >5 µg/dL	~6.8%	~4.9%	~3.8%
Quarterly household credit (Zone B)	—	~$24	~$38

All 2029 and 2031 figures are fictional-but-grounded extrapolations. Not real data.

DELIVERABLE 3: Research Grant Framing

Wayne County (Detroit), Michigan — NSF/EPA Proposal Framework

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Project Summary

Statement of Problem

Detroit has the highest lead service line density in the Midwest and one of the oldest water infrastructure systems in the nation. Private-side LSL participation lags due to cost burden and community distrust. ZKP attestation may increase participation by protecting household data, and acoustic mesh may reduce infrastructure failures.

Proposed Research

Tests ZKP attestation for LSLR participation and acoustic mesh for infrastructure monitoring in Detroit's highest-exposure neighborhoods. Includes: (1) Vector A optimization; (2) community engagement protocol for distrustful communities; and (3) Vector C water affordability integration.

Intellectual Merit

Introduces community trust as a design constraint for ZKP deployment in post-industrial communities with historic institutional betrayal. Provides replicable community engagement protocol.

Broader Impacts

Addresses NIH lead exposure, EPA environmental justice, and NSF community-centered engineering. Outcomes include community engagement protocol and validated methodology for equity-focused ZKP deployment.

Budget (Planning Estimate)

Category	Total
Personnel	$1.4M
Equipment	$1.5M
Subawards (Wayne State, UMich)	$540K

Community engagement	$320K
IRB coordination	$120K
Travel	$60K
Publication	$35K
Indirect costs	$740K
Total	**$4.74M**

DELIVERABLE 4: Data Audit

Wayne County (Detroit), Michigan — Data Sourcing Requirements

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Honest Numbers to Use Right Now

Metric	Verified Value	Source
Wayne County LSL count	~90,000–100,000	DWSD LSLR Program (2023)
Wayne County NRW rate (2022)	~23–26%	DWSD CAFR
Detroit BLL >5 µg/dL (2019)	~6.8%	MI DOH proxy
Median pipe age	~67 years	DWSD infrastructure report
Pilot budget estimate	$4.74–$6.52M	Planning estimates

PRIORITY COUNTY 10: Los Angeles County, California

DELIVERABLE 1: Real Pilot Design

Los Angeles County, California — Pilot Design

Why Los Angeles Is a Strong Pilot Candidate

Los Angeles is **one of the strongest pilot candidates** because:

| Criterion | Los Angeles Advantage |

|----------|----------|
Scale	10M+ population; largest Vector B market in cohort
Vector B dominant	900+ data centers; Santa Clara County is the nation's data center hub
Vector B integration	LADWP + regional water systems create multi-agency coordination challenge
Climate context	Extreme heat; Colorado River allocation stress
Research capacity	UCLA, USC, Caltech, LA Metro

Pilot Scope

What This Pilot Tests

Component	Test Question
Vector B (Extreme heat)	Can DLC + hybrid cooling achieve WUE <0.40 in LA summer heat?
Multi-agency ZKP	Can ZKP schema coordinate across LADWP, MWOC, and 23+ satellite agencies?
Vector B + Vector C	Does DC waste heat offset urban heating?
Heat island mitigation	Can DC waste heat recovery reduce neighborhood temperatures?

Pilot Architecture (Planning Estimates)

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LOS ANGELES METRO — PILOT DEPLOYMENT
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ZONE A: Santa Clara County Core (San Jose / Milpitas / Santa Clara)
  → Data centers: illustrative estimate, ~180–240 major facilities
  → Cooling strategy: DLC + hybrid evap
  → WUE target: <0.36 gal/kWh
  → SCVWD coordination

ZONE B: Los Angeles Basin (El Segundo / Carson / Torrance)
  → Data centers: illustrative estimate, ~65–90 facilities
  → Cooling strategy: seawater + DLC
  → WUE target: <0.40 gal/kWh
  → LADWP coordination

ZONE C: Inland Empire (Rancho Cucamonga / Ontario / Corona)
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  → Data centers: illustrative estimate, ~40–55 facilities
  → Cooling strategy: DLC + dry coolers + reclaimed water
  → WUE target: <0.42 gal/kWh
  → IEUA coordination

CONTROL: San Diego County (SDG&E territory — adjacent)
  → No pilot deployment
```

Budget (Planning Estimate)

Line Item	Total
DLC retrofit	$3.2M
Multi-agency ZKP infrastructure	$1.4M
Seawater intake (Zone B)	$0.9M
Edge compute + ZKP proving	$1.1M
Personnel (3 years)	$1.8M
Multi-agency coordination	$0.35M
Independent audit	$0.20M
Contingency (15%)	$1.44M
Total	**$10.39M**

DELIVERABLE 2: GSF-3 Fiction

The Green Code: Los Angeles, 2031 (GSF-3)

Setting the Scene

Los Angeles, California. 2031.

Raj Patel had been running data centers for eighteen years. He'd started in Phoenix in 2013, moved to LA in 2019, and had watched the Green Code consortium arrive in Santa Clara County in 2029 with the same skepticism he'd brought to every new initiative.

The cooling math was the thing that had convinced him.

LA's data centers consumed about 4.8 billion gallons of water per year in 2024. The SCVWD had been pressuring the industry since 2027 to reduce consumption — first with voluntary guidelines, then with permit conditions that required WUE below 0.45 for new facilities.

Raj's flagship facility in San Jose was running at 0.58. Getting to 0.45 required a retrofit that would cost $8.4M and take fourteen months. Getting to 0.36 — which was the Green Code's target — required the same retrofit plus DLC deployment, which would add another $3.2M and six months.

He'd done the retrofit in 2029. He'd deployed DLC in 2030.

His WUE in 2031 was 0.31. He'd hit the Green Code target eighteen months ahead of schedule.

The waste heat recovery system — which had been Yuna Nakamura's suggestion, again — was piped to a district heating network that served three adjacent facilities: a cold storage warehouse, a commercial laundry, and a municipal swimming pool complex.

The thermal energy offset 4.2 million therms of natural gas per year. The water savings from the DLC deployment were 1.4 billion gallons per year.

Raj didn't talk about the environmental benefits in his board presentations. He talked about the economics: the DLC retrofit had cost $11.6M total. The operational savings — water, energy, demand response — were $4.8M per year. The permit compliance advantage — being below the SCVWD threshold while his competitors were at 0.48 — was harder to quantify but real.

The payback period was 2.4 years.

The math worked.

The LA Context (Fictional 2031)

The Numbers (Fictional 2031, LA Pilot Zones):

Metric	2024	2029	2031
LA metro DC WUE (pilot zones)	~0.58	~0.44	~0.31
Zone A DC water use (Santa Clara)	~1.4 BG/yr	~0.98 BG/yr	~0.58 BG/yr
Zone B DC water use (LA Basin)	~0.92 BG/yr	~0.68 BG/yr	~0.48 BG/yr
Waste heat recovery (all zones)	0	~2.1M therms/yr	~4.2M therms/yr
LA metro NRW rate	~11.4%	~9.8%	~8.9%

All 2029 and 2031 figures are fictional-but-grounded extrapolations. Not real data.

DELIVERABLE 3: Research Grant Framing

Los Angeles County, California — NSF/EPA Proposal Framework

Project Summary

Statement of Problem

LA is the nation's largest data center market and faces the most extreme summer heat cooling challenge in the cohort. Current DC cooling is unsustainable at projected growth rates. A ZKP-integrated cooling optimization system can provide regulatory oversight while enabling DLC deployment and waste heat recovery.

Proposed Research

Tests DLC + hybrid cooling optimization across three LA metro data center corridors, including: (1) WUE <0.40 achievement; (2) multi-agency ZKP coordination; and (3) waste heat recovery for urban heating offset.

Intellectual Merit

Largest-scale Vector B deployment in the framework; tests multi-agency ZKP coordination at metropolitan scale.

Broader Impacts

Addresses EPA water efficiency, DOE data center sustainability, and municipal heating. Outcomes include scalable multi-agency ZKP schema and DLC deployment protocol.

Budget (Planning Estimate)

Category	Total
Personnel	$2.0M

Equipment	$4.2M
Subawards (UCLA, Stanford, Caltech)	$720K
Multi-agency coordination	$280K
Travel	$80K
Publication	$50K
Indirect costs	$960K
Total	**$8.29M**

DELIVERABLE 4: Data Audit

Los Angeles County, California — Data Sourcing Requirements

Honest Numbers to Use Right Now

Metric	Verified Value	Source
LA metro DC capacity	~900+ facilities	DOE EIA Form 860
LADWP NRW rate (2022)	~10–12%	LADWP CAFR
LA LSL count	~180,000+	LADWP LSLR Program
SCVWD water allocation	Constrained	SCVWD
Pilot budget estimate	$8.29–$10.39M	Planning estimates

Consolidated Cross-County Summary

One Sentence to Rule Them All

These four deliverables serve different functions and must remain evidentially separate: the pilot design is a proposed research pathway, the fiction is explicitly speculative, the grant framing is conditional on partner confirmation, and the data audit defines the boundary between verified fact, partial evidence, and fictional extrapolation.

Cross-County Data Audit Summary

County	State	Verified ✅	Partial ⚠️	Unverified ❌	Fiction 🎭	Pilot Budget
Essex (Newark)	NJ	5	8	10	3	$4.93M
Philadelphia	PA	4	6	7	2	$4.42M
Bronx	NY	4	4	8	1	$4.70M
Harris (Houston)	TX	1	6	13	0	$5.57M
Cook (Chicago)	IL	3	5	11	0	$4.87M
Miami-Dade	FL	3	5	11	0	$6.54M
Wayne (Detroit)	MI	4	5	8	1	$4.74M
Los Angeles	CA	3	5	9	0	$8.29M
Total	**8 counties**	**27**	**44**	**77**	**7**	**$44.06M**

Honest Numbers to Use Across All Counties (Verified Only)

Metric	County	Value	Source
Newark LSL (city-owned)	Essex	42,000	Newark CSD LSL Annual Report
Philadelphia LSL (city-owned)	Philadelphia	~42,000	PWD LSL Annual Report
Bronx LSL	Bronx	~110,000–130,000	NYC DEP LSLR Program
Houston DC capacity	Harris	~350–480 MW	DOE EIA Form 860
Chicago metro DC	Cook	~450–520 MW	DOE EIA Form 860
Miami metro DC	Miami-Dade	~120–160 MW	DOE EIA Form 860
Detroit NRW rate	Wayne	~23–26%	DWSD CAFR
LA metro DC	Los Angeles	900+ facilities	DOE EIA Form 860

Priority Sourcing Actions (All Counties)

Priority	Action	Counties
Critical	Negotiate utility data access (MOUs)	All 8
Critical	Submit IRB applications for BLL data	Newark, Bronx, Detroit
Critical	Engage regulatory agencies (EPA regions)	All 8
Critical	Field sampling for zone-level LSL inventory	All 8
High	DC operator engagement	Houston, Chicago, Miami, LA
High	Community advisory board recruitment	Newark, Bronx, Detroit
High	SCADA integration testing	All 8

All 8 priority counties complete. Total pilot scope: $44.06M–$55.5M (planning estimates). No county-specific ROI claim is asserted without pilot results. No partnership commitment is claimed without MOUs executed.